

SEOSAN, Korea, Republic of

WMO: 471290

Lat: 36.7767N

Lon: 126.4939E

Elev: 30

StdP: 100.96

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-10.0	-8.4	-16.3	0.9	-5.7	-14.7	1.1	-5.0	9.4	-1.7	8.2	-1.3	2.0	50	0.423

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	6.8	31.9	25.5	30.5	24.8	29.2	24.0	27.0	29.5	26.1	28.8	25.4	28.0	3.1	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
26.4	21.9	28.3	25.4	20.7	27.5	24.6	19.7	27.0	85.3	29.5	81.4	28.9	78.3	28.2	30.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
8.5	7.2	6.3	-13.0	33.7	2.1	1.4	-14.5	34.7	-15.8	35.6	-17.0	36.4	-18.5	37.4		
			-13.5	27.5	2.1	1.4	-15.1	28.5	-16.3	29.4	-17.6	30.2	-19.1	31.2		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	12.4	-1.6	0.2	4.9	11.1	16.8	21.6	24.9	25.7	21.0	14.7	7.6	0.9
	DBStd	9.94	3.53	3.56	3.39	3.51	2.62	1.90	2.07	2.25	2.52	3.26	3.92	3.72
	HDD10.0	1202	359	274	162	29	0	0	0	0	5	92	281	
	HDD18.3	2809	617	507	417	219	61	2	0	0	6	119	322	539
	CDD10.0	2072	0	1	4	61	212	347	463	486	330	150	20	0
	CDD18.3	637	0	0	0	1	14	99	204	227	86	6	0	0
	CDH23.3	4403	0	0	0	8	94	466	1441	1937	440	17	0	0
	CDH26.7	1109	0	0	0	1	8	68	366	602	65	0	0	0
Wind	WSAvg	2.4	2.3	2.5	3.0	3.0	2.8	2.3	2.7	2.3	2.0	1.9	2.2	2.3
Precipitation	PrecAvg	1199	30	33	46	85	97	112	261	236	148	61	57	34
	PrecMax	1788	88	89	116	210	223	439	508	534	375	292	164	80
	PrecMin	686	3	1	2	8	8	10	58	83	14	0	12	8
	PrecStd	242	22	24	29	54	55	98	104	117	92	71	40	18
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	10.2	13.9	17.7	24.5	27.7	30.1	34.1	34.0	29.9	25.0	19.8	12.9
		MCWB	7.3	9.3	11.2	14.6	17.4	20.5	25.4	26.5	22.5	18.1	14.3	9.0
	2%	DB	7.3	10.4	15.1	21.6	25.7	28.5	31.5	32.5	28.5	23.5	17.6	10.5
		MCWB	4.4	6.4	9.4	13.7	16.8	20.2	25.4	25.9	21.6	17.4	13.6	7.3
	5%	DB	5.5	8.2	13.1	19.8	24.1	27.2	30.0	31.2	27.2	22.0	15.9	8.7
		MCWB	3.0	4.7	8.3	12.6	16.3	20.2	25.2	25.5	21.3	16.4	12.1	6.0
	10%	DB	4.0	6.4	11.3	17.9	22.6	26.1	28.6	29.9	25.9	20.5	14.2	7.0
		MCWB	1.7	3.4	7.1	11.9	15.9	20.0	24.4	24.9	20.5	15.6	11.0	4.5
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	7.9	11.0	13.6	17.4	20.4	23.7	27.7	28.6	25.1	20.9	16.6	10.1
		MCDB	9.5	11.7	15.8	21.1	23.3	26.6	29.8	31.1	27.5	23.3	18.5	12.0
	2%	WB	5.1	7.0	11.1	15.6	19.1	22.7	26.6	27.4	23.9	19.1	14.6	8.0
		MCDB	6.9	9.4	13.6	19.3	22.6	25.5	29.1	30.1	26.5	21.4	16.7	9.9
	5%	WB	3.4	5.1	9.1	14.0	18.0	22.0	25.9	26.6	22.9	17.8	12.9	6.2
		MCDB	5.0	7.7	11.9	17.8	21.8	24.9	28.5	29.3	25.1	20.4	14.8	8.4
	10%	WB	1.9	3.7	7.6	12.7	17.0	21.2	25.2	25.9	21.9	16.7	11.5	4.8
		MCDB	3.8	6.1	10.6	16.4	20.7	24.1	27.8	28.6	24.3	19.4	13.7	7.0
Mean Daily Temperature Range	5% DB	MDBR	8.2	9.3	10.1	10.8	9.7	8.1	6.0	6.8	8.8	10.4	9.3	8.2
		MCDBR	10.0	12.1	12.9	13.5	12.5	10.6	8.3	8.7	10.0	11.4	11.3	10.4
		MCWBR	7.4	8.3	7.9	6.7	5.3	4.1	3.5	3.4	4.3	5.8	7.2	7.6
	5% WB	MCDBR	9.0	10.6	10.6	10.4	9.4	7.5	6.5	7.0	7.3	9.4	9.1	9.7
		MCWBR	7.3	8.2	7.6	6.4	5.3	3.8	3.4	3.7	4.2	6.1	7.1	7.8
Clear-Sky Solar Irradiance	taub	0.447	0.491	0.569	0.600	0.619	0.692	0.650	0.577	0.509	0.497	0.479	0.447	
	taud	1.937	1.861	1.721	1.700	1.714	1.638	1.775	1.953	2.060	1.955	1.942	1.953	
	Ebn at Noon	692	715	699	709	704	653	677	720	747	702	654	658	
	Edh at Noon	143	173	218	235	235	254	220	180	154	156	141	132	
All-Sky Solar Radiation	RadAvg	2.42	3.25	4.40	5.12	5.67	5.24	4.33	4.56	4.23	3.66	2.46	2.10	
	RadStd	0.16	0.28	0.43	0.38	0.56	0.56	0.66	0.66	0.44	0.28	0.26	0.15	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47)	Regional (42 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+93	+53

Nomenclature: See separate page